

GYUWON LEE

Email: gwl0711@gmail.com

RESEARCH INTERESTS

Brain-Computer Interfaces · Speech Neuroprostheses · Human Intracranial Electrophysiology · Neural Dynamics

EDUCATION

Seoul National University (SNU)

Seoul, Korea

M.S., Brain and Cognitive Sciences

Mar. 2023 – Feb. 2025

- GPA: 4.2/4.5 (97/100), Thesis advisor: Prof. Ki Woong Kim | Research mentor: Prof. Chun Kee Chung
- Thesis: “Improving Imagined Speech Decoding: Combining Adaptive Feature Selection with Linguistic Feature Decoding to Enhance Intelligibility”

Korea Advanced Institute of Science & Technology (KAIST)

Daejeon, Korea

B.S., Electrical Engineering

Mar. 2017 – Feb. 2023

- CGPA: 4.06/4.3 (97.6/100), Advisor: Prof. Young-gyu Yoon, Graduated **Summa Cum Laude**
- Thesis: “Calcium Imaging through Non-negative Matrix Factorization and Convolutional Sparse Coding: A Study on Neuron Signal Detection from Videos”

RESEARCH EXPERIENCE

KUH Neuro-Imaging Computational Analysis Lab (NICA)

Seoul, Korea

Research Assistant

Apr. 2025 – Present

- Developing learnable temporal-warping methods for speech-related MEG to uncover trial-specific neural timing variability and improve word decoding; temporal-alignment work accepted for presentation at SfN 2026.
- Developing a generative inverse framework to test whether high-frequency cortical activity obscured in scalp EEG can be recovered, using intracranial ECoG as a surrogate clean target, EEG-matched corruption models, and conditional diffusion posterior inference.
- Designed and directed an IRB-approved MEG/OPM speech-BCI study spanning audiovisual perception, overt speech, and imagined speech, integrating NI-DAQ and digital triggers for hardware-level synchronization.

SNU Human Brain Function Lab (HBF; PI: Prof. Chun Kee Chung)

Seoul, Korea

Project Coordinator, “The Alchemist Project”, Fully Implantable Closed-Loop BCI Consortium Mar. 2024 – Feb. 2025

- Coordinated a \$16.8M (KRW 21.8B) national neuroengineering consortium involving 13 PIs and 30 researchers across multiple institutions; coordinated planning and administration of SNU's ~\$0.8M (KRW 1.05B) annual project budget.
- Coordinated international data-sharing and speech-neuroprosthesis collaborations, including direct collaboration with researchers at the University of Melbourne.

Graduate Research Assistant, Brain and Cognitive Sciences Dept.

Mar. 2023 – Feb. 2025

- Led human iEEG speech-task recording sessions in 10 patients with drug-resistant epilepsy, integrating Blackrock splitter routing for simultaneous Natus clinical monitoring and Curry research acquisition.
- Designed and engineered a real-time imagined-speech decoding pipeline using Curry 9 TCP streaming with

online normalization and classification.

- Developed a cross-modal human iEEG-to-speech reconstruction framework using coupled neural and spectrogram autoencoders, latent-space alignment, and bidirectional cross-reconstruction objectives.
- Preprocessed tens of hours of human iEEG recordings for self-supervised neural representation learning with BrainBERT and HFO detection, with the HFO analyses contributing to a Nature Communications publication on human focal cortical dysplasia.

KAIST Neuro-Instrumentation & Computational Analysis Lab (NICA)

Daejeon, Korea

Research Assistant, Electrical Engineering Dept. (Advisor: Prof. Young-gyu Yoon)

Jan. 2022 – Dec. 2022

- Developed NMF and convolutional sparse-coding methods to decompose calcium-imaging videos into neuronal spatial components and temporal activity; B.S. thesis.

KAIST BCI Robotic Arm Control Project

Daejeon, Korea

Research Assistant

Jan. 2022 – Dec. 2022

- Led an EEG decoding team as a senior student; formulated a strategy and task for motor imagery decoding.

KAIST Biomedical Optics Lab (BMOL)

Daejeon, Korea

Research Assistant

Dec. 2017 – Feb. 2022

- Conducted optical diffraction tomography and neural-cell imaging experiments and contributed to planning optical-system improvements.

ADDITIONAL ENGINEERING EXPERIENCE

SN Academy

Seoul, Korea

AI Engineer / Co-Developer

Jul. 2025 – Present

- Engineered and deployed a full-stack personalized AI chat assistant that analyzes 24/7 user behavior logs and academic data to provide optimized guidance.
- Commercialized the platform, acquiring 300+ active paying subscribers and generating recurring revenue.

PUBLICATIONS

Park, Y. J., Kwon, J., **Lee, G.**, Meng, K., & Chung, C. K. (2026). "Spatiotemporal Dynamics in Prespeech Semantic Category Decoding: An Intracranial EEG Study". *eNeuro*, 13(4), ENEURO.0254-25.2026.

<https://doi.org/10.1523/ENEURO.0254-25.2026>

Lee, H., Hwang, J., Kim, K., **Lee, G.**, Chung, C.K., & Im, C. (2026). "Enhanced synthesis of passively heard speech from electrocorticography signals using image-to-image spectrogram translation". *Machine Learning with Applications*, 23, 100805. <https://doi.org/10.1016/j.mlwa.2025.100805>.

Kwon, J.*, **Lee, G.***, Park, Y. J., Lee, E. J., & Chung, C. K. (2025). "Mimed speech as an intermediary state between overt and imagined speech production in an electrocorticography study". *Scientific Reports*, 15, 33393. <https://doi.org/10.1038/s41598-025-18537-2>

**Equal contribution*

Cho, E., Kwon, J., **Lee, G.** et al. (2024). "Net Synaptic Drive of Fast-spiking Interneurons is Inverted towards Inhibition in Human FCD I Epilepsy". *Nature Communications*, 15, 6683.

<https://doi.org/10.1038/s41467-024-51065-7>

SELECTED CONFERENCE PRESENTATIONS

Lee, G. & Kim, J. S. (2026). "Temporal alignment improves MEG-based acoustic reconstruction during natural speech listening." Society for Neuroscience Annual Meeting (SfN 2026). Accepted poster presentation.

Lee, G. & Chung, C. K. (2024) "Decoding Number of Syllables from Human Intracranial Electroencephalography," Organization for Human Brain Mapping (OHBM), Seoul, Korea.

Lee, G., Kwon, J., Kim, K.W., & Chung, C. K. (2024) "Decoding Number of Syllables from Human Intracranial Electroencephalography," Cognition, Sleep, Mood, and Stress (COSMOS), Seongnam, Korea. ** *Excellent Presentation Award*.

Lee, G., Kwon, J., Kim, J.S., & Chung, C. K. (2023) "Decoding Phonemes with a Rule-based Restriction in Human Intracranial Electroencephalography," 26th Annual Meeting of the Korean Society for Brain and Neural Sciences (KSBNS), Busan, Korea.

Meng, K., Kwon, J., Park, Y.J., **Lee, G.**, Kim, J.S., Chung, C.K., & Grayden, D.B. (2023) "Decoding Intonation Patterns from Brain Signals for the Development of Speech Neuroprostheses," The Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), Sydney, Australia.

SELECTED PATENTS

Chung, C. K., Gwon, Y., Kwon, J., Roh, S., Park, Y. J., & **Lee, G.** (2025). "User-protective voice brain-computer interface device and method considering the intentionality of imagined speech, and non-transitory recording medium for performing the same." Korean Patent Application No. 10-2025-0148561 (pending).

Chung, C. K., Kwon, J., Park, Y. J., Gwon, Y., Roh, S., & **Lee, G.** (2025). "Method and apparatus for controlling objects using a context-integrated LLM based on decoded imagined-speech brain signals, and non-transitory recording medium for performing the same." Korean Patent Application No. 10-2025-0133662 (pending).

Chung, C. K., Kwon, J., Park, Y. J., & **Lee, G.** (2025), "Method and system for evaluating cognitive function based on brain waves", U.S. Patent Application No. 19/211,449 and Korean Patent Application No. 10-2024-0067565 (pending).

Chung, C. K., Kwon, J., Park, Y. J., & **Lee, G.** (2025), "User-customized language derivation method and device based on brainwave", U.S. Patent Application No. 19/188,098 and Korean Patent Application No. 10-2024-0068544 (pending).

TEACHING EXPERIENCE

Teaching Assistant, "Structure & Function of Human Brain," SNU Medicine Dept.

Mar. 2023 – Jun. 2023

Lecturer, "Classical Music Appreciation Course," KAIST (4 semesters)

Feb. 2018 – Dec. 2022

SCHOLARSHIPS

Brain Korea 21st Century (BK21) Scholarship, Seoul National University

Mar. 2023 – Feb. 2025

Full-tuition Scholarship, Korea Advanced Institute of Science & Technology

Mar. 2017 – Dec. 2022

One-year National Science & Engineering Scholarship, Korea Student Aid Foundation

2022

AWARDS AND HONORS

COSMOS (Cognition, Sleep, Mood, and Stress) Symposium

Seongnam, Korea

Excellence Poster Presentation Award

Jan. 2024

Korea Advanced Institute of Science and Technology

Daejeon, Korea

Dean's List

Fall 2017, Spring 2019

MILITARY EXPERIENCE

Air Force Operations Command (AFOC), Republic of Korea

Pyeongtaek, Korea

Satellite Communication Systems Operator-Maintainer, Squad Leader

Jun. 2020 – Dec. 2021

- Operated UHF/SHF satellite communication systems spanning ~300 lines and 40 stations; graduated with top honors from Satellite Operations Specialist Training.

PROFICIENCY IN SKILLS

Programming: Python (PyTorch, TensorFlow), MATLAB, C/C++, JavaScript/React

Neurophysiology & Experimental Tools: Curry 9, Natus, Profusion, STIM2, NI-DAQ

Software: Linux, Git/GitHub

Languages: Korean (native), English (advanced)